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Advances in Hazard Management for Machine Learning based Safety-Critical Systems

Operating in Unpredictable Environments

by

Haider M. Al-Shareefy

ORCID ID <https://orcid.org/0000-0002-8291-5620>

Thesis for the degree of Doctor of Philosophy

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Abstract

Faculty of Engineering and Physical Sciences

School of Electronics and Computer Science

Doctor of Philosophy

Appendix C Extended Literature Review

by

Haider M. Al-Shareefy

This Appendix is part of a PhD titled: Advances in Hazard Management for Machine Learning based Safety-Critical Systems Operating in Unpredictable Environments.

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Appendix C Extended Literature Review

C.1 General Systems Science Concepts

C.1.1 Complexity and Complicatedness

Reviewing the literature on complexity and complicatedness [1], the traditional systems engineering paradigm appears to have a vague distinction between complexity and complicatedness. The INCOSE [2] definition of complexity distinguishes between **simple**, **complicated**, and **complex systems** based on the nature of their elements and the relationships between those elements:

1. **Simple Systems:** Relationships between elements are easily observed and comprehended, leading to straightforward cause-and-effect understanding.
2. **Complicated Systems:** Relationships are intricate but can be unravelled and comprehended, enabling sufficient certainty about cause and effect. Traditional systems engineering (SE) methods work well here.
3. **Complex Systems:** Relationships between elements are woven together and not fully comprehended, leading to uncertainty about cause and effect. These systems often exhibit emergent behaviours, nonlinear dynamics, and counterintuitive outcomes.

As a holistic systems engineer, how do I use this definition objectively? There is no direct relationship to an already quantifiable aspect that can be re-adapted to solve real-world problems. Yes, there is an element of appreciation, but it is hard to translate such a definition in practice. This is because the definition does not tell me what a not-system is. INCOSE's definition tells me that all predictable organisations and unpredictable organisations of things are systems. But, I need a clear boundary of 1 and 0 so that I can say this is a bad choice because it leads to no system, and this is a good choice because it leads to the realisation of a system.

INCOSE does not directly define complicatedness and how it differs from complexity. So, based on INCOSE's explanation of complicated systems, **complicatedness** can be inferred as:

Complicatedness refers to the quality of a system having intricate but comprehensible relationships between its elements. These relationships can be systematically analysed, understood, and managed, leading to **sufficient certainty** in predicting cause-and-effect dynamics.

- **Key Characteristics of Complicatedness according to INCOSE:**

1. **Detailed Interconnectivity:** While the relationships among elements are numerous or detailed, they are not inherently unpredictable or emergent.
2. **Deterministic or Predictable Behaviour:** The system behaves according to fixed or known rules and patterns, allowing its behaviour to be unfolded and understood through analysis.
3. **Amenability to Reductionist Methods:** Problems can be broken down into smaller parts, solved individually, and recombined to form the overall solution.
4. **Finite State Space:** The system has a bounded and comprehensible range of possible states or outcomes.

Example: A **complicated system** like a traditional car engine has many interconnected parts (e.g., pistons, valves, fuel injectors), and their interactions can be intricate. However, engineers can understand, predict, and manage these interactions using well-established principles and methods.

According to the INCOSE characterisation of Simple and Complex, why is a predictable car engine not simple? Why is it not Complex? Would an autonomous car be a complex or complicated system? What would you class a domino puzzle of 100 pieces and 10^9 pieces?

In summary, INCOSE appears to draw the following distinction:

Complicatedness:

- Defined by manageable intricacy.
- Relationships are fixed and predictable.
- Reductionist approaches (e.g., decomposition, reassembly) work well.

Complexity:

- Defined by emergent behaviours.
- Relationships are interdependent and non-linear.
- Requires holistic, iterative, and systems thinking approaches.

At the same time, INCOSE distinguishes the following properties of emergence in general:

- **Emergence is a System-wide Property:** Systems exhibit emergent properties that derive from their elements and interactions but cannot be reduced to those elements [3].
- **Universal Phenomenon:** Emergence is recognised as a fundamental property of all systems [4].
- **Distinct from Elemental Properties:** Emergence arises from the "properties the system has, but the elements themselves do not" [5].

Interestingly, the cited reference in [4], Sillito et.al, postulate the following:

“Emergence, the appearance of a new phenomenon or capability as a result of relation or interaction between objects, is key in differentiating between objects that are systems and those that are not.”

In their remarks, they implicitly state that objects that do not exhibit emergent behaviours are not systems. It is interesting that the authors consider the feasibility of “non-system” to exist. So, according to INCOSE, complicated systems do not exhibit emergent properties. And according to Sillito et al., complicated systems are not systems. This is the problem we found in the literature, which is confusion and ambiguity.

So what? Why is understanding the difference between complexity and complicatedness important for our systems approach and for any aspiring systems engineer working on safety-critical autonomous systems?

Simply because Compounded Epistemic Uncertainty Problems (problems that involve autonomous systems, people, traditionally engineered systems, environment, etc.) in a safety-critical application are confusingly complicated issues and hard to achieve robust certainty (robust trustworthy safety case) that can withstand the test of time, in comparison to the issues that involve engineering deterministic safety-critical CPS. Understanding the nature of complicatedness is the key aspect of reducing epistemic uncertainty for Compounded Epistemic Uncertainty Problems.

This section is covered in more detail in our conference paper, which outlines the core concepts of complexity and complicatedness in the following work [6]. We will briefly explain our prior work as the foundation for the

C.2 Architect’s epistemic uncertainty due to semantic gap

Let’s take an example. Suppose the architect has engineered a dataset. The architect must demonstrate that their large, engineered dataset faithfully captures the required behaviour in given scenarios. This indicates that the claimed epistemic uncertainty is deliberately targeted and reduced. For example, the architect may claim that although the epistemic uncertainty curve of the chosen dataset distribution strategy suggests a non-linear behaviour, tests over unforeseeable validation datasets demonstrated superior performance. The dataset completely covers the requirements. How can the architect show that written requirements are truly captured in those 213,000 images? Mainly including them in a limited safety case. How can the

quality assurance team, a regulator, or a court (in case of a catastrophic failure) check all 213,000 images and validate that requirement are truly captured?

There is a significant risk that machine learning datasets for developing ai-based computer vision systems may need to capture or represent the necessary safety and security requirements fully. These datasets are crucial in shaping the behaviour and decision-making capabilities of the AI system. If key safety or security aspects are not included or properly represented during the dataset development, the resulting system may behave unpredictably in real-world scenarios. This can lead to vulnerabilities, including susceptibility to adversarial attacks, biased decision-making, or unsafe operations, compromising the system's reliability, ethical standards, and user safety.

The main challenge of this risk lies in the following premise:

It is difficult to close the semantic gap [7] [8] between the written description of a requirement (for example, safety) and the infinite possible ways they can appear in a pictorial form.

In deterministic systems like circuit boards, every component is precisely defined and engineered based on established physical and mechanical properties. Consider the following:

- **Specificity of component characteristics:** for example, based on circuit requirements, resistors can be chosen with exact resistance, tolerance, and power ratings. Capacitors, inductors, and transistors are similarly selected and arranged to achieve the desired behaviours.
- **Trustable predictability:** the arrangement and interaction of these components follow clear principles, allowing predictable and consistent behaviour under known conditions.
- **Trustable implementation of risk mitigation:** engineers can systematically identify and address risks such as electrical interference, overheating, or component failure using rigorous testing and simulation tools. And demonstrate their capture in the design.

The deterministic nature of such systems enables easier trust over their design and performance, making it easier to systematically meet safety and security requirements.

Challenges in machine learning dataset design

In contrast, machine learning systems rely on abstract and often subjective components to be engineered, such as visual representations of target objects of interest (tois), for their design. For example:

- **The problem of specificity: abstract representation of acceptable inputs.** Unlike a physical resistor with well-defined physical properties, a visual representation of a resistor in an image dataset is far less deterministic. Variability in size, orientation, lighting, background, and partial occlusions complicates ensuring consistent identification by an AI system. From a requirement engineering perspective, such a problem poses an intellectual burden in specifying a visual representation of a physical object in a way that can be unambiguous and thoroughly tested. If we select a 10-ohm resistor, we can entirely and unambiguously test such a property using a voltmeter. How can we visually specify a complete representation of a resistor to test a dataset of images to have a resistor in them completely?
- **The problem of trustable implementation of risk mitigation: semantic gap between requirements and captured acceptable inputs.** It is particularly challenging to bridge the gap between textual descriptions of requirements (e.g., "identify humans in all scenarios") and their infinite visual manifestations.
 - **Example 1:** identifying a human in an image raises questions such as:
 - Does a full-body image qualify as a human representation, or is a hand sufficient?
 - Should the system still classify it as a human if a person is partially visible behind a barrier (e.g., with only a leg or arm showing)?
 - **Example 2:** how should the system handle scenarios where a billboard or other object visually overlaps with a human? If we include it in Multiple images, we technically identify a completely non-human object as part of a human. How will this bias the dataset? What if the billboard has unethical gratification on it? Or images of an animal or another car? Now, we have introduced less-priority objects in terms of safety into the image of a human.
 - Why would such a scenario be important? An autonomous car must accurately differentiate between a human and a billboard. If a billboard visually overlaps a partially visible human, the system must not ignore the human presence, as doing so could result in a dangerous collision.

ASE challenge 3: it is very hard to predict, trace, and risk-manage the share size of infinite acceptable input scenarios associated with an open, complicated environment interacting with an autonomous system and vice versa.

What do we mean by “*infinite acceptable inputs*”?

Let's take a simple light switch as an example; the acceptable inputs of a light switch are an electrical current and a mechanical input that is pressed either on or off. A car engine is more

complex than a light switch; it accepts a wider range of inputs. Nonetheless, the set of all possible acceptable inputs is specifiable and finite in both cases and can be measured precisely. However, when considering a perception system run over a neural network, which is tasked with detecting a flying drone, for example, there are infinite ways that a flying drone may visually appear as input. Each possible way is classed as an acceptable input. Therefore, systems engineers must specify infinite possible inputs in a traceable manner that cannot be precisely measured.

Therefore, in some high-performance cars, we can develop a far more trustable assurance case for a highly complicated car engine than we could for a neural network with no moving parts. This says something about our understanding of what simplicity and complexity mean and the role of the problem-solver's ability to predict.

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